Advanced Image Processing

Class Hours: 11:30 am – 1:00 pm (T, Th)

Class: EE 308 (EE Building)

Instructors

Soma Biswas (EE)
somabiswas@iisc.ac.in

Rajiv Soundararajan (ECE)
rajivs@iisc.ac.in

Indian Institute of Science, Bangalore.

Syllabus

- **Image Features** - Scale Invariant Feature Transform (SIFT), Speeded Up Robust Features (SURF), Generalized Hough Transform, Texture features, Evaluation
- **Deep Learning of image features** - Convolution layers, fully connected layers, back propagation, non-linear activation, batch normalization, network architectures
- **Image Segmentation** - Graph-based techniques, Active Shape Models, Active Appearance Models, deep learning based segmentation
- **Image Restoration** - Image denoising using statistical priors, order statistics based filters, deep learning methods, deblurring and sharpening, defogging, low light image enhancement
- **Image Compression** - Entropy coding, lossless JPEG, quantization, DCT, JPEG
- **Object detection** – RCNN and its improved variants like Fast-RCNN, Faster RCNN, SSD
- **Instance Segmentation** - Mask-RCNN
- **Image Quality** - Natural scene statistics, quality assessment based on structural and statistical approaches, blind quality assessment
- **Generative Models**- Generative adversarial networks, variational auto encoders, style transfer, cycle GAN, diffusion models.
- **Vision Transformers**

Grading

- Assignments (4) – 35%
- Midterm – 15%
- Final Exam – 25%
- Final Project – 25%

Pre-requisite: Digital Image Processing

Team Name: E9 246 Advanced Image Processing 2025

Code: 35ivor6

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Recently, my team visited one of the premier institutions in India for campus hiring. We were disappointed to find that students pursuing Masters in Signal Processing had opted for ML courses over Signal Processing, resulting in a lack of knowledge in their core field. Many students struggled to answer basic questions about filters and transforms.

Nowadays, profiles across various branches (CS/Mech/Meta/Civil/...) appear similar, with everyone highlighting ML projects using Python. This raises the question: where is the core domain knowledge that sets candidates apart? While the job market influences this trend, not all jobs require only ML skills. ML should complement your core domain expertise.

For students interested in joining my group in Multimedia Systems at Qualcomm, we expect knowledge in Embedded Systems, Processor Architecture, OS concepts, Signal Processing, Image/Video/Audio/Speech processing, Computer Vision, C/C++ programming, and ML. There are numerous resources available to learn these concepts, and I encourage you to explore them.

To professors and students, please focus on core domain knowledge alongside ML to create differentiation.

BTW, during that campus hiring drive, we offered a position to a student who had both Signal Processing and ML knowledge!

Leave your thoughts here...